Detecting Humanness across Social Media Platforms

MDI Scholar: Rebecca Ansell, MS CS, 2025
Research Team: *The Humanness Project,* Dr. Lisa Singh, Dr. Leticia Bode, Dr. Tiago Ventura

Abstract

Humanness refers to the degree to which a piece of content is perceived to be generated by a human rather than by an artificial intelligence (AI). This research investigates how well humans can distinguish AI generated posts, from human authors, and the effectiveness of various embeddings and neural network architectures for classifying humanness using social media data from YouTube and X. We evaluated multiple models with different embeddings. The performance was assessed using relevant metrics. Our findings highlight the potential of specific embeddings and network configurations in predicting humanness with moderate accuracy, enabling insights into the interpretability of perceived humanness in social media posts.

Introduction

In the ever-evolving landscape of social media and generative artificial intelligence, distinguishing between human-written content and artificial intelligence (AI) generated text has become increasingly important. This study focuses on YouTube comments and X posts related to the 2024 election cycle.

Objectives:

- Analyze the perceived humanness of these posts using human annotators by using Bradley-Terry and ELO rankings to determine the most human-like posts.
- Building models to detect perceived humanness.

Research Questions:

1. Are humans able to detect AI-generated posts?
2. Are machine learning models able to detect AI-generated posts?
3. How well can we create models to detect perceived humanness?

This approach allows for an exploration of AI's influence on social media public's ability to discern human from non-human textual contributions.

Acknowledgements

Thank you to Dr. Lisa Singh, Dr. Leticia Bode, Dr. Tiago Ventura, and Rebecca Vanarsdall for your guidance, teaching, and ideas. Thank you to Kenny Ahmed for help with data collection, and to the REU students for data labelling. Thank you to the Massive Data Institute and the McCourt School for giving me the opportunity to work on a project like this. Thank you to the Tech and Public Policy Grants Program for supporting this work.

Methodology

1. Data Collection:
- Human Posts Data
  - Collected from X and YouTube through keywords
- Generated Posts Data
  - Generated from ChatGPT 4o and 3.5 using the same keywords

2. Data Labelling:
- Annotators were given pairwise comparisons of different post combinations and labelled which sounded more like AI
- Way to measure perceived humanness

3. Preprocessing:
- Data cleaning:
  - General formatting, censoring usernames and hyperlinks
- Ranking the posts
  - Using Bradley-Terry and ELO to rank posts from most human-like to most AI-like

4. Model Building
- Created multiple models with different embedding strategies

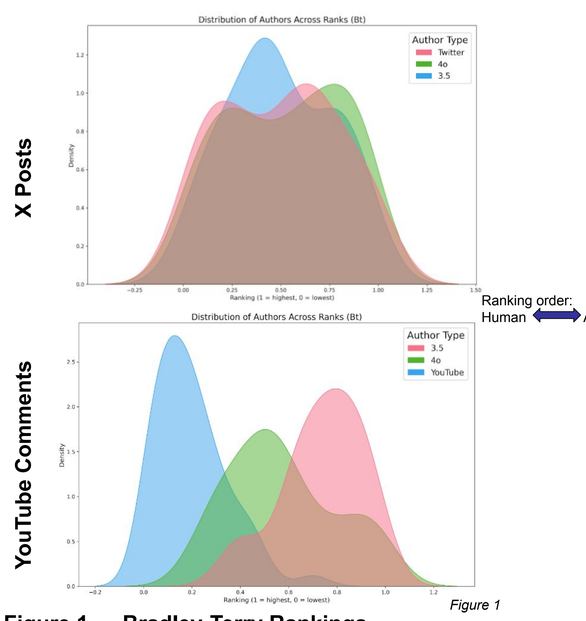


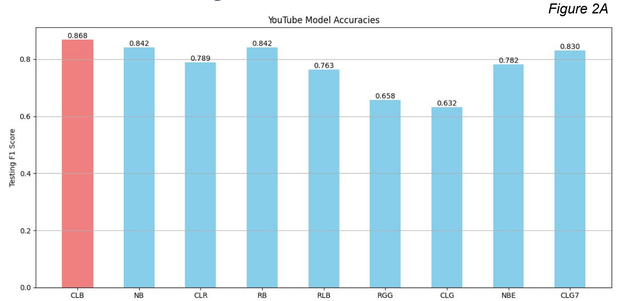
Figure 1

Figure 1 — Bradley-Terry Rankings
Using the BT Model, posts have been ranked based off of their pairwise comparisons results from most perceived as human to most perceived as AI. We can see that for X the posts are very difficult to distinguish, whereas YouTube appears to be well discerned by annotators. All pairs were double-labelled and there was a 49% disagreement between annotators.

Example of an X labelling task:
Which of the posts (Post A or Post B) is more likely to have been generated by an AI Model?
A: "Ensuring election integrity is crucial for democracy. #Democracy"
B: "I thought mail in voting was all fraudulent? If it's rigged, why do we vote?"

Preliminary Results

Figure 2A



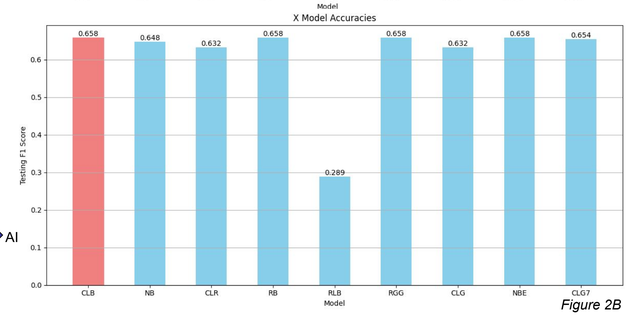


Figure 2B

Abbreviation	Model
CLB	CNN with LSTM (BERT)
NB	Perceptron NN (BERT)
CLR	CNN with LSTM (RoBERTa)
RB	RNN (BERT)
RLB	RNN with LSTM (BERT)
RGG	RNN with GRU layer (GPT-2)
CLG	CNN with LSTM (GPT-2)
NBE	Perceptron NN (BERT) (ELO ranking)
CLG7	CNN with LSTM (gemma-7b)

Figure 2 — Classifier Performance
2A YouTube: This figure shows the Testing F1 scores of various models applied to YouTube data.
Figure 2B: This figure shows the Testing F1 scores of various models applied to X data.

Discussion:

From the density graphs, We can see that for X the posts are very difficult to distinguish, whereas YouTube appears to be well discerned by annotators. The best model for both YouTube comments and X posts combines BERT embeddings with CNN for local pattern recognition and LSTM for sequential context. This model performed well because it effectively captures both local features and long-term dependencies in the text, leveraging the powerful contextual embeddings provided by BERT.

Limitations

- Limited labelled data covering election topics that are constantly changing.
- Human annotators had a difficult time differentiating between human and AI-generated content from X, this has limited our model building so far.
- We could not generate misinformation from any other chat bots except for GPT models, a greater diversity of chatbot sources would allow us to have a more diverse set of data.
- The election candidates have changed since the original data collection, we will need to expand our data collection and labeling to account for the change in candidates. We will also need to collect data on new topics that have emerged since the original data collection phase.

Ethical Considerations

- We censored all usernames and hyperlinks for the labeling surveys to protect user privacy.
- We purposefully generated misinformation to emulate real X posts and YouTube comments for our research study only.

Future Directions

1. We will use these models on election 2024 data to assess relative levels of humanness on different platforms.
2. Expand data collection to include the new democratic party's presidential nominee.
3. Expand study to other platforms: Reddit, TruthSocial.
4. Generate data from a wider variety of chatbots.
5. Further fine tuning and hyperparameter grid searching of current models.
6. Using more advanced embedding strategies such as Llama3.1, gemma-27b, and bloom-560m.

References

1. Carlson, David, and Jacob M. Montgomery. "A Pairwise Comparison Framework for Fast, Flexible, and Reliable Human Coding of Political Texts." *American Political Science Review*, vol. 111, no. 4, 2017, pp. 835-843, https://doi.org/10.1017/S0003055417000302.
2. Jia, Chenyan, and Taeyoung Lee. "Journalistic Interventions Matter: Understanding How Americans Perceive Fact-Checking Labels." *Harvard Kennedy School (HKS) Misinformation Review*, vol. 5, no. 2, 2024, https://doi.org/10.37016/mr-2020-138.
3. Routh, Jennifer, et al. "The Use of Pairwise Comparisons and the Elo Algorithm to Quantitatively Rate a Large Number of Items in a Single Dimension." EdArXiv, 16 Dec. 2022. Web.